E Linear Regression: Replacing Data with Increasing Sample Size

In the framework of Mobahi et al. (2020) and Dohmatob et al. (2024a), we consider sequences of linear models fit to the previous model's synthetic outputs. Within this framework, Dohmatob et al. (2024a) proved that if data are replaced with each model fitting iteration and the training data cardinality remains constant, then the test squared error scales linearly with the number of model fitting iterations n:

$$E_{\text{test}}^{\text{Replace}}(\hat{w}_n) = \frac{\sigma^2 d}{T-d-1} \times n \tag{9}$$

In this work, we lightly adapt the argument of Dohmatob et al. (2024a) to study the effects if data accumulate with each model fitting iteration. We specifically considered the case where the training data cardinality increases by a constant T with each model-fitting iteration i.e. the ith model is fit using $T \times i$ data, where T data are "real" and then each subsequently fit model contributes its own T synthetic data to the accumulating data. In this setting, the test squared error is upper bounded independent of the number of iterations.

$$E_{\text{test}}^{\text{Accumulate}}(\hat{w}_n) = \frac{\sigma^2 d}{T-d-1} \times \sum_{k=1}^{n} \frac{1}{k^2} \leq \frac{\sigma^2 d}{T-d-1} \times \frac{\pi^2}{6} \tag{10}$$

In the main text, we focus on the replace and accumulate data settings because prior work focused on replacing data and we wished to study how accumulating data affects model collapse. However, a much richer landscape of outcomes is possible. For instance, and as pointed out in personal correspondence with Dohmatob et al. (2024a), one can consider what we term the "Replace-Multiple" setting, in which one fits the i-th linear model using $T \times i$ data sampled from the $(i-1)$-th linear model. Replace-Multiple is a useful baseline for Accumulate because it matches the amount of training data at each model fitting iteration. Under Replace-Multiple, the test squared error grows logarithmically:

$$E_{\text{test}}^{\text{Replace-Multiple}}(\hat{w}_n) = \frac{\sigma^2 d}{T-d-1} \times \sum_{k=1}^{n} \frac{1}{k} \approx \frac{\sigma^2 d}{T-d-1} \times \log(n) \tag{11}$$

Replace-Multiple has the drawback of not matching the total amount of compute of Accumulate since each iteration of Replace-Multiple draws $T \times i$ samples from the most recent model, whereas Accumulate draws T samples from the most recent model. Other baselines are also possible, but we leave these to future work. We focus on accumulating data as we feel real and synthetic data are likely to accumulate in the real world as time progresses.

F Additional Linear Regression Numerical Results

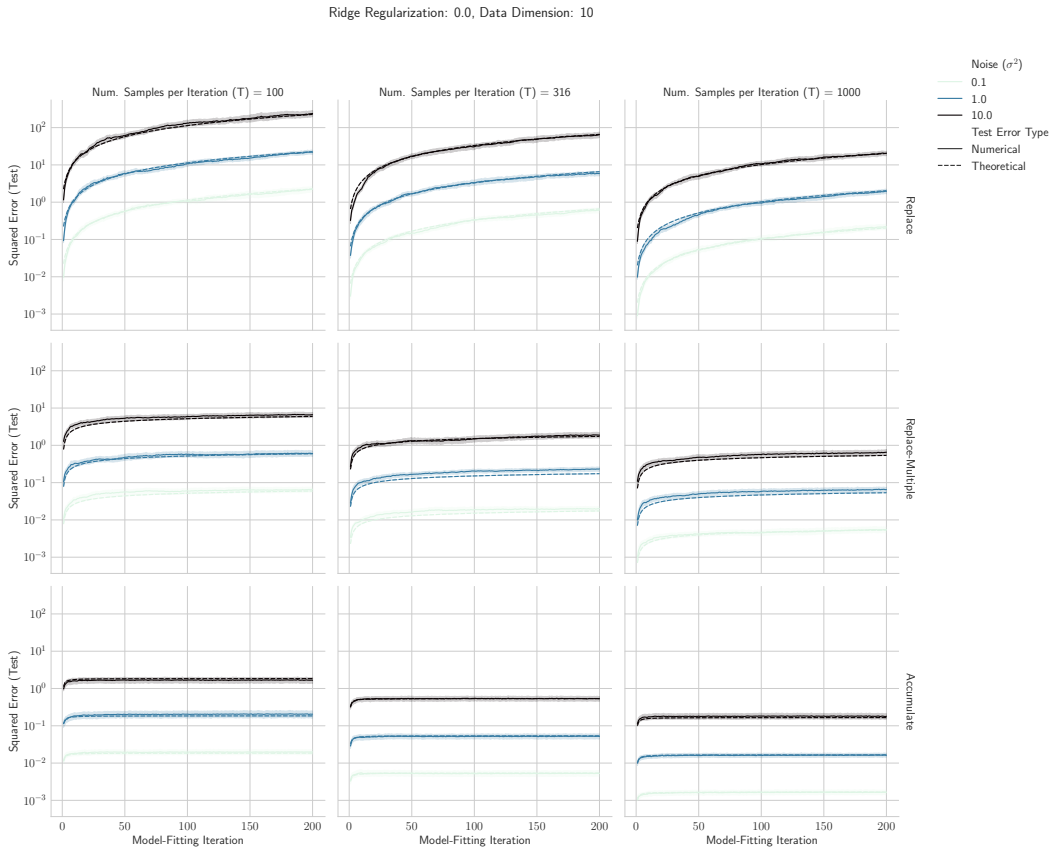

Figure 16: **Accumulating data across iterations avoids model collapse in linear regression.** We consider sequences of linear models recurrently fit to generated targets by previous iterations of models. Replace (Top): If each linear model is fit to the generated targets of *only* the preceding linear model i.e. data are replaced, then the test squared error grows linearly with the number of model-fitting iterations iterations n. Replace-Multiple (Middle): If each linear model is fit to $T \times i$ samples from the $(i-1)$-th model (i.e. the same amount of data as Accumulate), then the test squared error grows logarithmically with the number of model-fitting iterations; see Appendix E for more details. Accumulate (Bottom): If each linear model is instead fit to the generate targets of *all* the preceding linear models i.e. data accumulate, then the test squared error has a finite upper bound, independent of the number of iterations. This suggests that data accumulation might be a robust solution for mitigating model collapse. This figure is similar to Figure 7 but displaying *log* test squared error and more model-fitting iterations for additional clarity.